

Title

Efficacy Comparison of Regional Anesthesia Techniques Following Cesarean Delivery

Abstract

Sixty postpartum women undergoing cesarean delivery were randomized to receive either an erector spinae plane block (n = 30) or a transversus abdominis plane block (n = 30) for postoperative analgesia. Pain severity, analgesia duration, and satisfaction were compared between groups. The erector spinae plane block provided analgesia for a mean of 16 hours compared to 14 hours with the transversus abdominis plane block (not statistically significant, $p = 0.32$). Pain scores one hour postoperatively were lower in the erector spinae plane group (mean 2.1 vs 3.2 on 10-point scale, $p = 0.03$). However, no significant differences were observed at subsequent timepoints (4, 8, 12, 24 hours). Requirement for rescue analgesia was lower in the erector spinae plane group (mean 0 vs 1 additional dose, $p = 0.045$). Patient satisfaction with pain management was high in both groups (mean satisfaction 8.4/10 both groups). Postpartum depression screening scores at 4-6 weeks postpartum were similar between groups (no significant difference). Both regional anesthesia techniques provided effective postoperative pain control; the erector spinae plane block showed marginal advantage in early pain scores and reduced additional analgesia requirement, though clinical significance of these differences is limited.